TorchJD: Jacobian Descent in PyTorch

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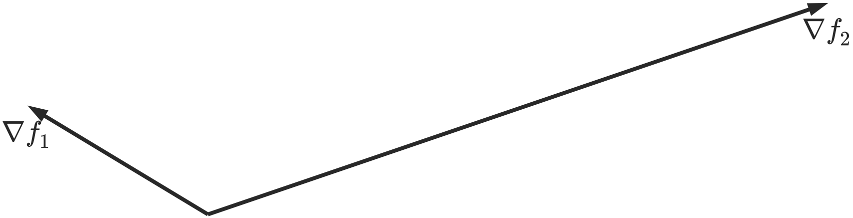
GitHub
github.com/SimplexLab/TorchJD

TorchJD
torchjd.org

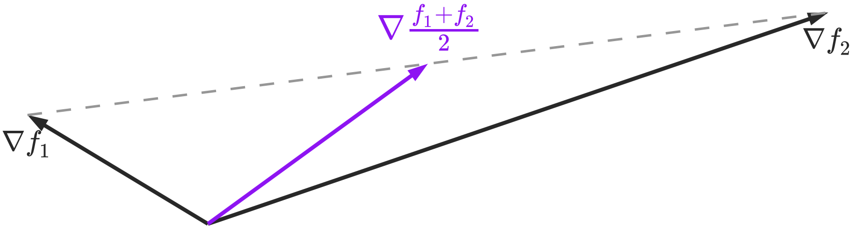
arXiv
arxiv.org/pdf/2406.16232

Motivation

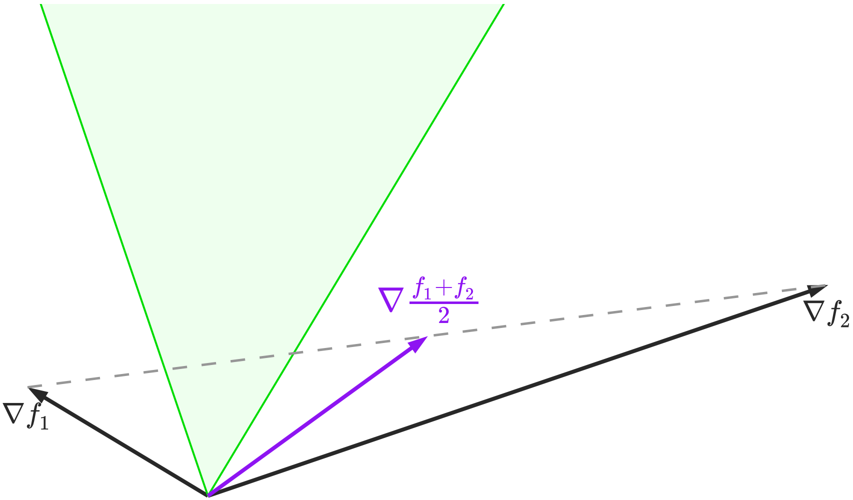
Jacobian descent intuition



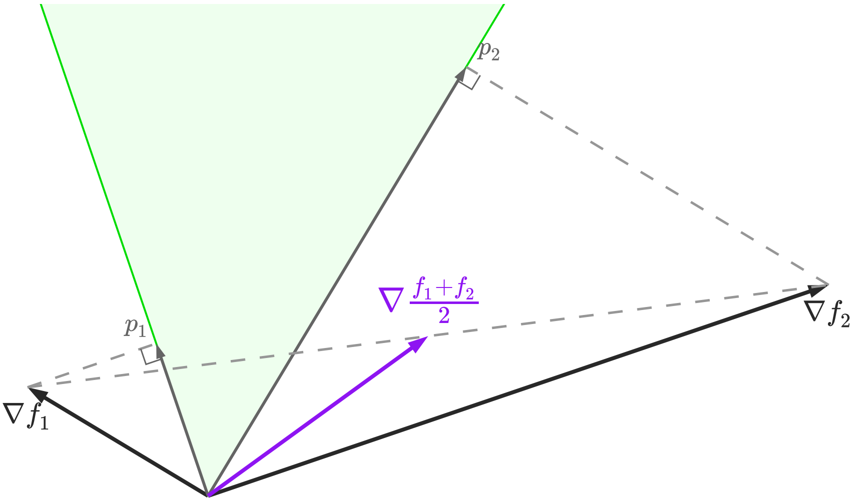
Jacobian descent intuition



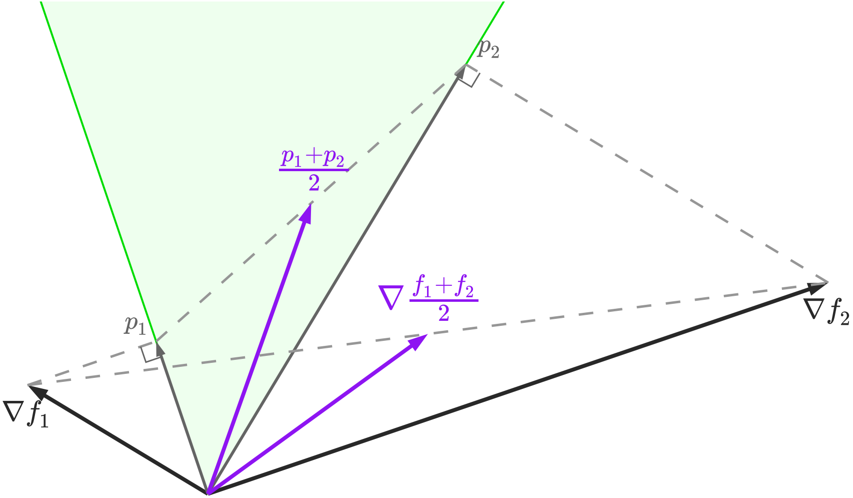
Jacobian descent intuition



Jacobian descent intuition



Jacobian descent intuition



TorchJD: core functions

Library	Function	Description
`torch`	`autograd.grad`	compute gradients
`torch`	`autograd.backward`	compute gradients and accumulate in `.grad`
`torchjd`	`autojac.jac`	compute jacobians
`torchjd`	`autojac.backward`	compute jacobians and accumulate in `.jac`
`torchjd`	`autojac.jac_to_grad`	aggregate `.jac` fields into `.grad`

TorchJD: basic usage

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

batch_size = 16
model = Linear(10, 1)

for x, y in dataloader:
  z = model(x).squeeze(dim=1)  # shape: [16]
  losses = loss_fn(z, y)  # shape: [16]
  autojac.backward(losses)
  # parameters now have a .jac field
  autojac.jac_to_grad(model.parameters(), UPGrad())
  # parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

More examples at torchjd.org.

Discussion

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Can JD solve problems that GD can't?